MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : MC 401/MC-401/MC401 Environmental Sciences

UPID : 004421

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What are causes of acid rain and what is its effects on natural environment ?

(II) What do you understand by GCM ?

(III) What are In-situ and Ex-situ conservation method of biodiversity ?

(IV) Graphically explain the relationship between BOD and DO.

(V) What are the various sources of water pollution ?

(VI) Mention name of few gases generated from Incineration plant .

(VII) Mention various types of noise pollution and its effects on human being.

(VIII) What authorizes the Central Government of India in Environmental Protection Act, 1986 ?

(IX) What are the various types of earthquake waves generated in the earth surface?

(X) Mention the types and function of various components of ecosystem.

(XI) Mention the various elements of air pollution as per NAAQS.

(XII) What is meant by 5-day BOD and Ultimate BOD?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. (i) Explain with neat diagram the internal structure of the Earth. [5]
(ii) What do you mean by rock cycle?

3. (i) What do you mean by maximum mixing depth? [5]
(ii) Differentiate between Ambient lapse rate and Adiabatic lapse rate.

4. (i) Define flood and how can it be mitigated and controlled ? [5]
(ii) What do you understand by anthropogenic environmental degradation ?

5. (i) Write the equation for population growth and explain the notations. [5]
(ii) How to calculate "Natural Increase Rate" ?

6. (i) How arsenic get dissolved in groundwater in the aquifer system ? [5]
(iii) How many districts in West Bengal are exposed to groundwater arsenic contamination ?

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (i) What are different types of solid wastes ? [4 + 6 + 5]
(ii) What are the methods of solid waste disposal and management ?
(iii) What are the advantages and disadvantages of Land filling of solid waste ?

8. (i) What are Nutrient cycles and how are they important for our environment ? [4 + 6 + 5]
(ii) Draw a flowchart for Nitrogen Cycle.
(ii) Explain Biogeochemical cycle by a diagram .

9. (i) Mention various types of greenhouse gases and its effects on terrestrial ecosystems. [6 + 5 + 4]
(ii) How global warming and climate change are related?
(iii) What is short wave and long wave radiation ?

10. (i) Why sustainable growth is considered modern concept of development ? [5 + 3 + 7]
(ii) How population growth in India can be controlled ?
(iii) If the present population of the World is 8.00 billion, what would be the population of the World after 25 years with a growth rate of 1% per year ?

11. (i) What is a groundwater aquifer, aquiclude and aquitard. Explain with neat diagram. [6 + 4 + 5]
(ii) How groundwater is recharged ?

(iv) What do you understand by Rooftop rainwater harvesting ?

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC- CS401/PCC-CS401/PCCCS 401/PCCCS401 Discrete Mathematics

UPID : 004407

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) Explain the difference between a permutation and a combination, and provide an example of each.

(II) What is the truth value of ¬(P∧Q)¬(P ∧Q) if P is true and Q is false?

(III) What is an abelian group?

(IV) What is the chromatic number of a graph?

(V) What is composition of mapping?

(VI) If you have 6 different colors of socks, how many ways can you choose 2 pairs (4 socks total)?

(VII) What is the truth value of (P∧¬Q)∨(Q∧¬P)(P ∧¬Q)∨(Q∧¬P) if P and Q are both true?

(VIII) Define a group in abstract algebra and explain the group axioms.

(IX) What is the minimum number of edges required for a connected graph with n vertices to be a tree?

(X) Define the union of two sets.

(XI) In a city, there are 30 traffic lights. If each traffic light can be either red, yellow, or green, what can you guarantee about the colors of some traffic lights?

(XII) If p→(q∧r) and r→s are both true, what can you conclude about p→s?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Prove by mathematical induction 3n<n! for all positive integers n≥6 [5]
3. A drawer contains ten black and ten white socks. You reach in and pull some out without looking at them. What is the least number of socks you must pull out to be sure to get a matched pair? Explain how the answer follows from the pigeonhole principle. [5]
4. What is De Morgan's Law in propositional logic? Provide both forms of the law. [5]
5. In a ring R if $x^3 = x$ for all x ∈ R, then show that R is commutative. [5]
6. Consider a connected undirected graph G with n vertices and m edges. Prove that if G has no cycles, then m must be less than n. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Let G be a group in which for some integer n > 1, $(ab)^n = a^n b^n$ for all a, b ∈ G. Show that [5]
   a) $G^n = \{ x^n \mid x \in G \}$ is a normal subgroup of G.
   b) $G^{n-1} = \{ x^{n-1} \mid x \in G \}$ is a normal subgroup of G.
   (b) If ϕ : G → H is a homomorphism and G is abelian, then Imϕ = { ϕ(g) | g ∈ G } is abelian. [5]
   (c) Prove that any group of order 15 is cyclic. [5]
8. (a) prove that (p→q)∧(q→r)→(p→r) is a tautology [5]
   (b) Without truth table find the DNF and CNF of the expression [10]
   ~(p ∨ q) ↔ (p ∧ q)
9. (a) State and prove Division Algorithm [8]
   (b) Determine the Recursive formula for the sequance 3,9,21..........Also find the 11th term [7]
10. (a) By principle of inclusion and exclusion find the number of 10 combinations of the elements of the set S containing 5 a's, 4 b's 5 c's and 7 d's [8]
    (b) State the first form of Pigeonhole Principle and the prove that in a party where guests are handshaking among themselves.there will always be at least two guests who have shaken hands the same number of times. [7]

11. (a) Consider the following directed weighted graph G with vertices V={A,B,C,D,E} and edges with their corresponding weights: [8]

(A,B,5)
(A,C,3)
(B,C,2)
(B,D,6)
(C,D,7)
(C,E,4)
(D,E,5)

Apply Dijkstra's shortest path algorithm to find the shortest paths from vertex AA to all other vertices in the graph.

(b) Prove that a simple graph with n vertices and k components can have atmost 1/2(n-k)(n-k+1) edges [7]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC-CS402 Computer Architecture

UPID : 004442

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is meant by data dependence?
(II) Which page replacement algorithm suffers from Belady's anomaly?
(III) Which architecture is/are suitable for realizing SIMD?
(IV) What is meant by Branch Prediction?
(V) The throughput of a super scalar processor is ___________ .
(VI) Which unit is responsible for translation of logical address to physical address?
(VII) The ______ plays a very vital role in case of super scalar processors.
(VIII) The set of loosely connected computers are called as _____.
(IX) Write the equation for Amdahl's Law.
(X) Write the statement for memory inclusion property.
(XI) What is meant by instruction level parallelism?
(XII) In tightly coupled systems, the microprocessors share _____________.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Given a 4 segment pipeline whereby each segment has a delay time as follows: [5]
Segment1: 40 ns Segment2: 25 ns Segment3: 45 ns Segment4: 45 ns
The delay time for the interface register is 5ns. Calculate the:
i) cycle time of the non-pipeline and pipeline,
ii) execution time for 100 tasks,
iii) real speedup,
iv) maximum speed up.

3. In a simple machine with load-store architecture having clock rate of 1.8GHz and the following specifications: [5]

Operations	Frequency	Number of Clock Cycles
ALU	40%	1
Load	20%	2
Store	10%	2
Branch	30%	2

Calculate CPI and MIPS rating for the machine.

4. What is a superscalar processor? State the advantages of vector computer. [5]
5. State the differences between static network and dynamic network. Explain the hypercube interconnection with n=3. [5]
6. Compare superscalar, super-pipelined and superscalar-super-pipelined architecture. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. a. Compare tightly coupled system and loosely coupled system. [6+9]
b. Explain with suitable diagram: multiprocessor architectures (UMA, NUMA, COMA).

8. a. What is page fault? [2+9+4]
b. Given page reference string: [1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6]. Compare the page fault rates for LRU, FIFO and Optimal page replacement algorithm.
c. Discuss the implementation of virtual memory through segmentation with suitable diagram.

9. a. Discuss SIMD array processor architecture with suitable diagram. [5+2+2+6]
   b. What are vector stride and vectorization?
   c. What is the difference between scalar processor and vector processor?
   d. Explain vector gather and scatter instructions with suitable diagrams.

10. a. What is cache coherency? [2+3+6+4]
    b. Explain the MESI protocol briefly.
    c. Explain the snoopy bus protocol for cache coherency.
    d. Explain how synchronization is ensured in multiprocessor environment?

11. a. Explain VLIW architectures with suitable diagram. [6+5+4]
    b. State the advantages and disadvantages of VLIW architecture.
    c. What are the hurdles in superscalar architecture?

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PCC-CS 403/PCC-CS403/PCC-CSBS401/PCCCS403 Formal Language & Automata Theory

UPID : 004423

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What type of grammar is Context-sensitive grammar ?

(II) Give two applications of TM .

(III) What is a universal Turing machine?

(IV) Give two applications of Context free languages .

(V) Give two examples/applications designed as finite state systems.

(VI) What is context-free language?

(VII) Define context sensitive grammar.

(VIII) How can you proceed for Turing machine construction?

(IX) What are the various representation of TM?

(X) Define finite automaton.

(XI) What do you mean by k-equivalent states ?

(XII) When a grammar is said to be ambiguous ?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Find context-free grammars for the language (with $n \geq 0$, $m \geq 0$, $k \geq 0$): [5]
$L = \{a^n b^m c^k, k = n + 2m\}$.

3. Discuss Universal Turing Machine (TM) [5]

4. Show that the following grammar is ambiguous: [5]
$S \rightarrow aSbS \mid bSaS \mid \lambda$.

5. Consider the grammar G=({S},{a,b},S,P) [5]
with P given by
S->aSb
S->∈
What will be the language of the grammar?

6. What is the Turing Machine Halting Problem? [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. Discuss with examples how to simplify context-free grammar: [5+5+5]
a) By removing the useless production
b) By moving λ –production
c) By removing unit production

8. (a) What is Turing Machine? [5]
(b) How can Turing Machines be used as language acceptors? [3]
(c) For Σ = {0, 1}, design a Turing machine that accepts the language denoted by the regular expression 00*. [7]

9. (a) When a Problem is said to be Undecidable ? [5]
(b) Let Σ = {a, b}.Write down the grammar that generates the following language : [5]
$L = \{a^n b^m : n \geq 0, m < n\}$.
(c) Show that the grammars [5]
$S \rightarrow aSb \mid bSa \mid SS \mid a$

and

$S \rightarrow aSb|bSa|a$

are not equivalent.

10. (a) Consider the language $L = \{a^n : n = 3 \text{ or } n \text{ is even}\}$. Construct a DFA for this language. [5]

    (b) For $\Sigma = \{a, b\}$, construct a DFA that accepts all strings with an even number of a's. [5]

    (c) For $\Sigma = \{a, b\}$, construct DFA that accepts all strings with exactly one a. [5]

11. Discuss Chomsky hierarchy of languages with examples. [15]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : ES-CS401/PCC-CS 404/PCC-CS404/PCC-CSD 402/PCCCS404 Design & Analysis of Algorithms

UPID : 004416

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the appropriate data structure for Prim's Minimum Spanning Tree algorithm?

(II) What is represented by the asymptotic notation O(1) ?

(III) Time complexity for recurrence relation T(n)=2 T(n/2) + n is _________________.

(IV) The notation Ω(n) is the formal way to express the _______________ bound of an algorithm's running time.

(V) Fractional knapsack problem is solved most efficiently by which algorithm design technique ?

(VI) What is the appropriate data structure for Depth First Search algorithm?

(VII) Which algorithm is used to solve a maximum flow problem?

(VIII) How many solutions are there for 8 queens problem on 8*8 board?

(IX) Time complexity of Heap sort with n items is ________________.

(X) The 0-1 Knapsack problem can be solved using Greedy algorithm -state True or False.

(XI) Which algorithm is used to solve the single source shortest path problem in a graph with negative edge weights?

(XII) Given items as {value,weight} pairs {{40,20},{30,10},{20,5}}. The capacity of knapsack is 20. Find the maximum value output assuming items to be divisible.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What are the differences between Dijkstra's algorithm and Bellman-Ford algorithm? Compare their time complexities. [5]

3. Analyze the time complexity of quick sort algorithm for best case and worst case using recurrence relations. [5]

4. Write the Floyd-Warshal Algorithm for all-pairs shortest path problem. What will be the time complexity? [5]

5. For the given set of 5 items and the knapsack capacity of 10 kg, find the maximum profit. You are allowed to take fractional amount of any item. [5]
Weights, Wi = {3, 3, 2, 5, 1}
Profits, Pi = {10, 15, 10, 12, 8}

6. Write short note on Randomized algorithms. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Find the minimum number of multiplications required for the following matrix chain multiplication using Dynamic programming: A (10 x 20) * B (20 x 50) * C (50 x 1) * D (1 x 100) [7]

(b) Write the algorithm for single source shortest path (-ve edge weights allowed) [5]

(c) Discuss the advantages and drawbacks of Dynamic Programming. [3]

8. (a) What are the features of any Algorithm? [3]

(b) Discuss the different Asymptotic notations and their importance. [5]

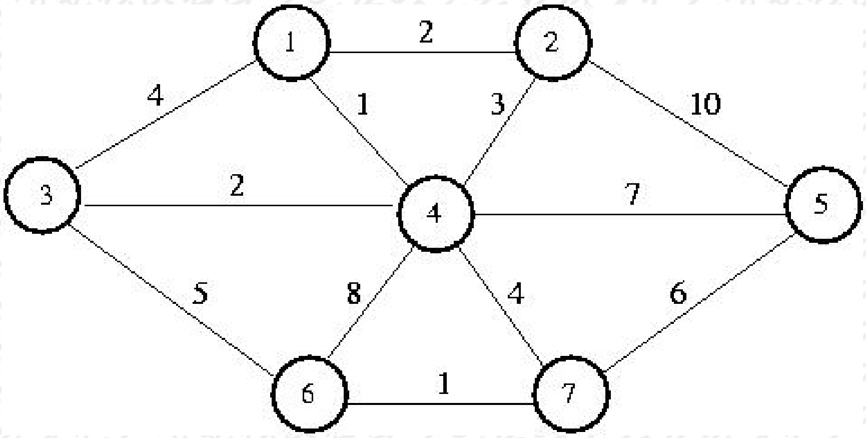
(c) Write the recurrence relation for computing the time complexity of "Towers of Hanoi" problem. Also solve the recurrence relation to get the time complexity. [7]

9. (a) Write the greedy algorithm for job sequencing with deadline. [4]

(b) Using greedy method ,find an optimal solution to the problem of job sequencing with deadline where n=4, (p1,p2,p3,p4)=(100,10,5,27) and (d1,d2,d3,d4)=(2,1,2,1). [5]

(c) Define spanning tree. [2]

(d) Using Prim's algorithm generate the MST from the following graph [4]



10. (a) Write and explain the classes P, NP, NP-hard and NP-complete. [7]
    (b) What is reducibility? [3]
    (c) "If any NP-complete problem can be solved in polynomial time then P=NP " explain. [5]
11. (a) What is a Flow Network? What are the constraints of flow in a network? [5]
    (b) What is an augmenting path in a flow network? Explain residual capacity - how it is calculated. [5]
    (c) Write the Ford-Fulkerson Algorithm for finding maximum flow. [5]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-B401/BSC 401/BSC-401/BSC401 Biology

UPID : 004408

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) The portion of the growth curve where rapid growth of bacteria is observed is known as ____________

(II) The human eye can focus objects at different distances by adjusting the focal length of the eye lens. This is due to__________.

(III) What is meant by the term osmoregulation?

(IV) Who is known as father of genetics?

(V) What is meant by power of accommodation of the eye ?

(VI) Enzyme increases the rate of reaction by lowering the activation energy. Is this statement is true or false?

(VII) What is cistron?

(VIII) State two economically important uses of heterotrophic bacteria.

(IX) What is the main function of kinase?

(X) The growth of bacterial population follows a geometric progression.
a) True
b) False

(XI) Are viruses living or non-living?

(XII) Fluid Thioglycollate medium is used for the cultivation of which type of organism?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. How are co-factors different from prosthetic groups? [5]
3. Give characteristics of genetic code. [5]
4. Differentiate between DNA/RNA. [5]
5. Explain Krebs cycle. draw suitable flowchart for explanation. [5]
6. The sequence of the coding strand of DNA in a transcription unit is mentioned below. [5]
3′ AATGCAGCTATTAGG 5′
Write the sequence for:
1. Its complementary strand
2. Its mRNA

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. Illustrate the two models by which an enzyme holds the substrate. [15]
8. Explain steps of Gycolysis in details. [15]
9. Give functions of Proteins as receptors and structural elements. [15]
10. Describe the characteristics of the individuals with the following chromosomal abnormalities: [15]
Trisomy at chromosome 21
XXY
XO
11. A tall plant with red flowers (dominant) is crossed with a dwarf plant with white flowers (recessive). Work out a dihybrid cross and state the dihybrid ratio. What will be the effect on the dihybrid ratio if the two genes are interacting with each other? [15]

*** END OF PAPER ***